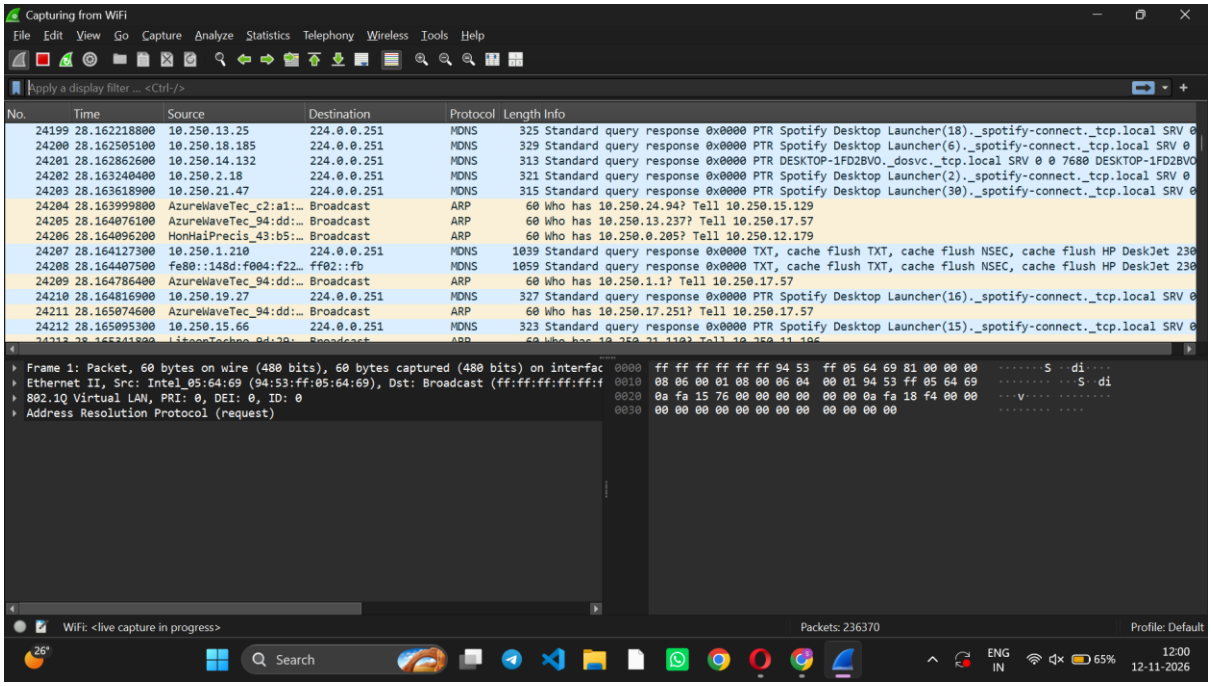
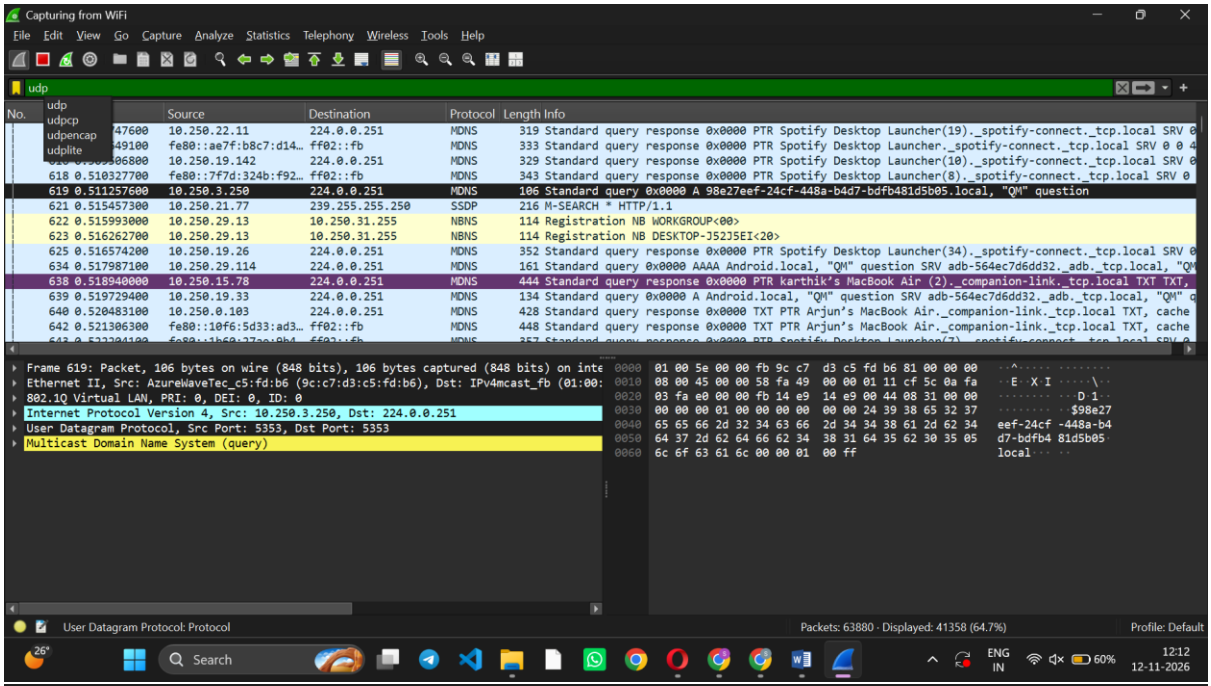
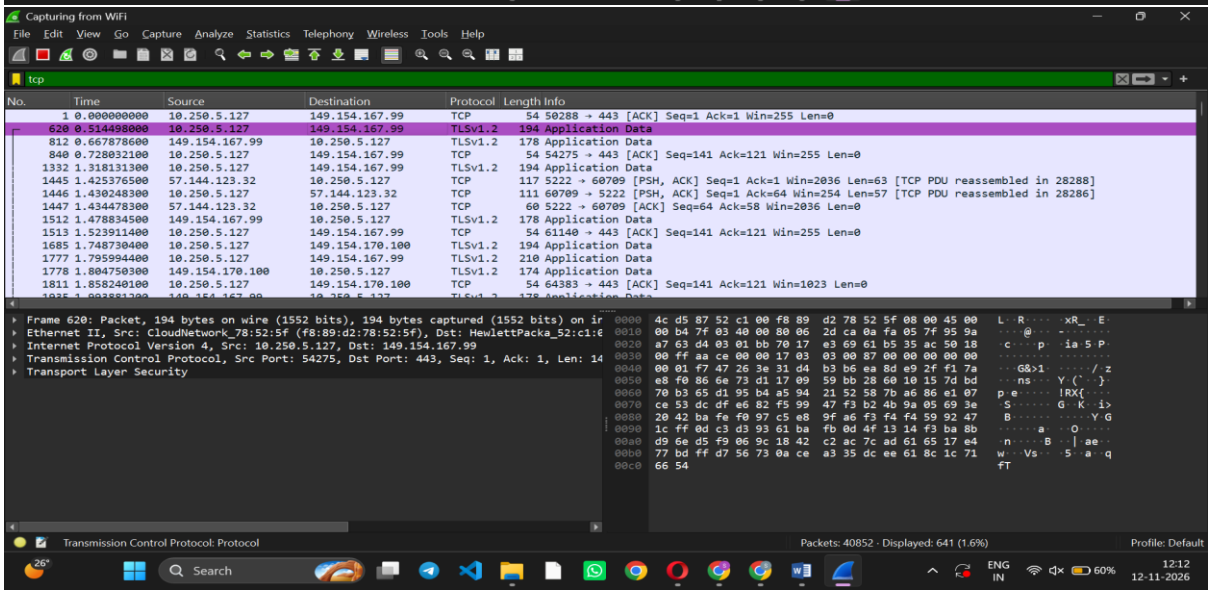
2520090140 – U.Vinay Sampath

WEEK 1 – SKILL




Capturing from WiFi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Length Info
47600 10.250.22.11 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(19)._spotify-connect._tcp.local SRV 0
49100 fe80::ae7f:b8c7:d14… ff02::fb MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 4
806800 10.250.19.142 224.0.0.251 MDNS 329 Standard query response 0x0000 PTR Spotify Desktop Launcher(10)._spotify-connect._tcp.local SRV 0
618 0.510327700 fe80::7f7d:324b:f92… ff02::fb MDNS 343 Standard query response 0x0000 PTR Spotify Desktop Launcher(8)._spotify-connect._tcp.local SRV 0
619 0.511257600 10.250.3.250 224.0.0.251 MDNS 106 Standard query 0x0000 A 98e27eef-24cf-448a-b4d7-bdfb481d5b05.local, "QM" question
621 0.515457300 10.250.21.77 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
622 0.515993000 10.250.29.13 10.250.31.255 NBNS 114 Registration NB WORKGROUP<00>
623 0.516262700 10.250.29.13 10.250.31.255 NBNS 114 Registration NB DESKTOP-J52J5EI<20>
625 0.516574200 10.250.19.26 224.0.0.251 MDNS 352 Standard query response 0x0000 PTR Spotify Desktop Launcher(34)._spotify-connect._tcp.local SRV 0
634 0.517987100 10.250.29.114 224.0.0.251 MDNS 161 Standard query 0x0000 AAAA Android.local, "QM" question SRV adb-564ec7d6dd32._adb._tcp.local, "QM
638 0.518940000 10.250.15.78 224.0.0.251 MDNS 444 Standard query response 0x0000 PTR karthik's MacBook Air (2)._companion-link._tcp.local TXT TXT,
639 0.519729400 10.250.19.33 224.0.0.251 MDNS 134 Standard query 0x0000 A Android.local, "QM" question SRV adb-564ec7d6dd32._adb._tcp.local, "QM"
640 0.520483100 10.250.0.103 224.0.0.251 MDNS 428 Standard query response 0x0000 TXT PTR Arjun's MacBook Air._companion-link._tcp.local TXT, cache
642 0.521306300 fe80::10f6:5d33:ad3… ff02::fb MDNS 448 Standard query response 0x0000 TXT PTR Arjun's MacBook Air._companion-link._tcp.local TXT, cache
Frame 619: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on inte
Ethernet II, Src: AzureWaveTec_c5:fd:b6 (9c:c7:d3:c5:fd:b6), Dst: IPv4mcast_fb (01:00:
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
Internet Protocol Version 4, Src: 10.250.3.250, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
0000 01 00 5e 00 00 fb 9c c7 d3 c5 fd b6 81 00 00 00
0010 08 00 45 00 00 58 fa 49 00 00 01 11 cf 5c 0a fa
0020 03 fa e0 00 00 fb 14 e9 14 e9 00 44 08 31 00 00
0030 00 00 00 01 00 00 00 00 00 00 24 39 38 65 32 37
0040 65 65 66 2d 32 34 63 66 2d 34 34 38 61 2d 62 34
0050 64 37 2d 62 64 66 62 34 38 31 64 35 62 30 35 05
0060 6c 6f 63 61 6c 00 00 01 00 ff
User Datagram Protocol: Protocol
Packets: 63880 · Displayed: 41358 (64.7%)
Profile: Default
Search
ENG IN
60%
12:12 12-11-2026


Capturing from WiFi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
1 0.000000000 10.250.5.127 149.154.167.99 TCP 54 50288 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=0
620 0.514498600 10.250.5.127 149.154.167.99 TLSv1.2 194 Application Data
812 0.667878600 149.154.167.99 10.250.5.127 TLSv1.2 178 Application Data
840 0.728032100 10.250.5.127 149.154.167.99 TCP 54 54275 → 443 [ACK] Seq=141 Ack=121 Win=255 Len=0
1332 1.318131300 10.250.5.127 149.154.167.99 TLSv1.2 194 Application Data
1445 1.425376500 57.144.123.32 10.250.5.127 TCP 117 5222 → 60709 [PSH, ACK] Seq=1 Ack=1 Win=2036 Len=63 [TCP PDU reassembled in 28288]
1446 1.430248300 10.250.5.127 57.144.123.32 TCP 111 60709 → 5222 [PSH, ACK] Seq=1 Ack=64 Win=254 Len=57 [TCP PDU reassembled in 28286]
1447 1.434478300 57.144.123.32 10.250.5.127 TCP 60 5222 → 60709 [ACK] Seq=64 Ack=58 Win=2036 Len=0
1512 1.478834500 149.154.167.99 10.250.5.127 TLSv1.2 178 Application Data
1513 1.523911400 10.250.5.127 149.154.167.99 TCP 54 61140 → 443 [ACK] Seq=141 Ack=121 Win=255 Len=0
1685 1.748730400 10.250.5.127 149.154.170.100 TLSv1.2 194 Application Data
1777 1.795994400 10.250.5.127 149.154.167.99 TLSv1.2 210 Application Data
1778 1.804750300 149.154.170.100 10.250.5.127 TLSv1.2 174 Application Data
1811 1.858240100 10.250.5.127 149.154.170.100 TCP 54 64383 → 443 [ACK] Seq=141 Ack=121 Win=1023 Len=0
Frame 620: Packet, 194 bytes on wire (1552 bits), 194 bytes captured (1552 bits) on ir
Ethernet II, Src: CloudNetwork_78:52:5f (f8:89:d2:78:52:5f), Dst: HewlettPacka_52:c1:6
Internet Protocol Version 4, Src: 10.250.5.127, Dst: 149.154.167.99
Transmission Control Protocol, Src Port: 54275, Dst Port: 443, Seq: 1, Ack: 1, Len: 14
Transport Layer Security
0000 4c d5 87 52 c1 00 f8 89 d2 78 52 5f 08 00 45 00
0010 00 b4 7f 03 40 00 80 06 2d ca 0a fa 05 7f 95 9a
0020 a7 63 d4 03 01 bb 70 17 e3 69 61 b5 35 ac 50 18
0030 00 ff aa ce 00 00 17 03 03 00 87 00 00 00 00 00
0040 00 01 f7 47 26 3e 31 d4 b3 b6 ea 8d e9 2f f1 7a
0050 e8 f0 86 6a 73 d1 17 09 59 bb 28 60 10 15 7d bd
0060 70 b3 65 d1 95 b4 a5 94 21 52 58 7b a6 86 e1 07
0070 ce 53 dc df e6 82 f5 99 47 f3 b2 4b 9a 05 69 3e
0080 20 42 ba fe f0 97 c5 e8 9f a6 f3 f4 f4 59 92 47
0090 1c ff 0d c3 d3 93 61 ba fb 0d 4f 13 14 f3 ba 8b
00a0 d9 6e d5 f9 06 9c 18 42 c2 ac 7c ad 61 65 17 e4
00b0 77 bd ff d7 56 73 0a ce a3 35 dc ee 61 8c 1c 71
00c0 66 54
Transmission Control Protocol: Protocol
Packets: 40852 · Displayed: 641 (1.6%)
Profile: Default
Search
ENG IN
60%
12:12 12-11-2026

Parameter	TCP Packet	UDP Packet
Frame Length	533 bytes	106 bytes
Ethernet Header Length	14 bytes	14 bytes (+4B VLAN)
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	519 bytes	88 bytes
Transport Header Length	20 bytes	8 bytes
Payload Length	479 bytes	60 bytes

1. **What is encapsulation?**
   Encapsulation is the process where each protocol layer wraps application data with its own control information (header, and sometimes a trailer) as the data moves down the network stack before transmission.
2. **What is the purpose of the Ethernet header?**
   The Ethernet header (Layer 2) enables local network delivery by specifying the source and destination MAC addresses and identifying the encapsulated upper-layer protocol (such as IPv4 or ARP) via the EtherType field.
3. **What information is stored in the IP header?**
   The IP header (Layer 3) stores logical addressing and routing details, including the source and destination IP addresses, IP version, header length, total packet length, Time to Live (TTL), protocol number (e.g., TCP or UDP), fragmentation flags, and a header checksum.

4. What is the difference between TCP and UDP headers?

- **TCP header:** Typically 20–60 bytes long; includes sequence numbers, acknowledgment numbers, window size, and control flags (SYN, ACK, FIN) to guarantee reliable, ordered, connection-oriented delivery.
- **UDP header:** Fixed at 8 bytes; contains only source port, destination port, length, and checksum, offering fast, connectionless, best-effort delivery with minimal overhead.

5.Why does each protocol add a header?

Each protocol layer adds a header to provide the specific addressing, routing, error checking, sequencing, and delivery instructions needed for its layer to function independently and hand off data properly at the receiving device.

6. What is packet overhead? Packet overhead is the total number of extra control bytes (headers and trailers) added across protocol layers that do not carry actual user or application payload data.

7. How does header overhead affect network efficiency? Header overhead consumes bandwidth and transmission capacity without carrying user data, lowering payload efficiency ($\frac{\text{Payload}}{\text{Total Frame Size}} \times 100$). Efficiency drops significantly when sending many small payloads because the fixed header bytes make up a large percentage of each frame.